Cover Page

- **Problem Statement/Concept Title :** Smart Vote – A Secure and Transparent Blockchain-Based Online Voting System
- **Theme :** Blockchain & Secure Digital Governance
- **Team Name (Registered on portal) :** Smart Vote
- **State/UT :** Tamil Nadu

Smart Vote

Problem – Proposed Solution

Core problem: Inaccessible and untrusted voting – need for a secure, transparent online voting system. **Proposed solution:** Smart Vote, a blockchain-based remote voting platform.

Detailed Explanation of Solution

- Remote voting through web / mobile platforms
- Aadhaar-based login + facial recognition (DeepFace)
- Votes digitally signed, SHA-256 hashed and stored as blocks on a shared ledger
- Stack: Python Flask + MySQL + Web3 smart contracts

How It Addresses the Problem

- **Access:** vote from anywhere; multilingual & voice-guided UI
- **Identity:** Aadhaar + face check, one voter – one vote
- **Tampering:** immutable blocks expose any change
- **Trust:** public, verifiable results

Innovation & Uniqueness

- Smart contract checks signature, blocks duplicates, adds noise for anonymity
- Hash-only storage keeps the vote private
- Anyone can audit via blockchain explorer
- Offline registration + contingency plan

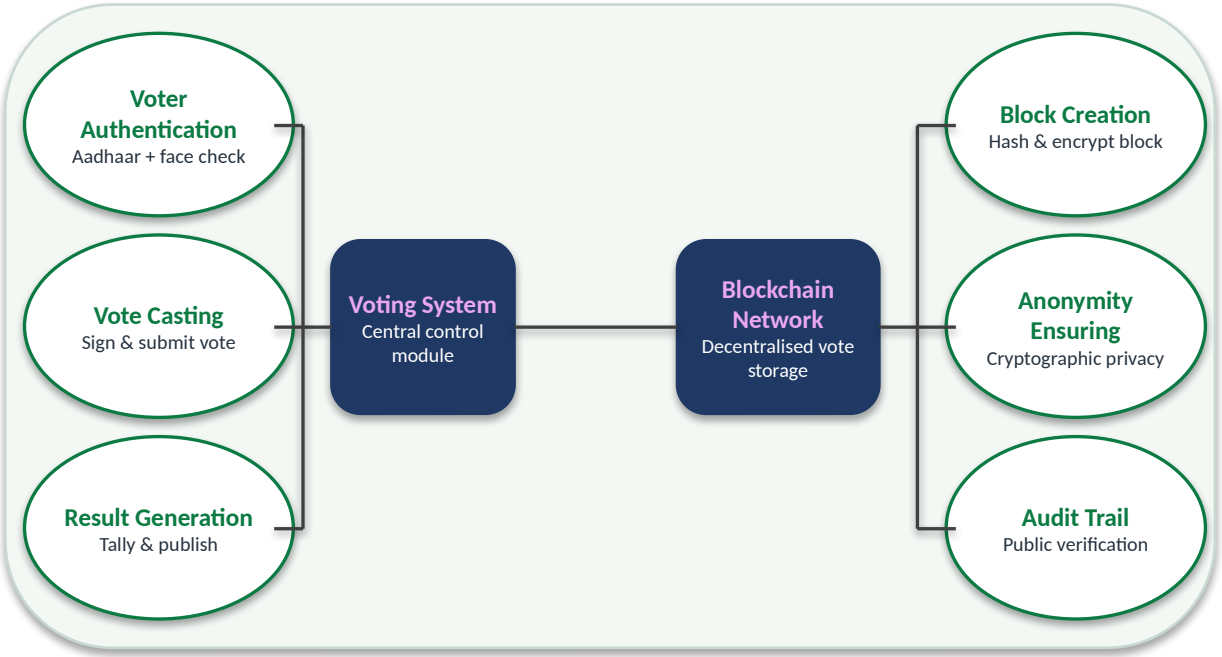
Voting workflow

1 Register & key generation → 2 Aadhaar + face authentication → 3 Sign & cast vote → 4 Smart-contract verification → 5 Hash & store in block → 6 Public audit & results

Smart Vote

Technical Approach

System Architecture



Technologies Used

- Python (Flask)
- MySQL
- Web3 · Smart Contracts
- DeepFace
- SHA-256 / 512
- End-to-end encryption

Methodology & Implementation Flow

1. **Registration & Key Generation**
   Voter creates public–private key pair (Web3 wallet); authority stores public key + voter ID
2. **Smart Contract Set-up**
   Authority deploys contract with election parameters and eligible-voter list
3. **Vote Casting**
   Voter signs the vote with private key; submitted as a Web3 transaction
4. **Verification & Privacy**
   Contract validates signature, blocks duplicates, adds controlled noise
5. **Immutable Storage & Audit**
   Validated vote stored on-chain; final data published for public audit

GitHub / GitLab: [Add repository link]

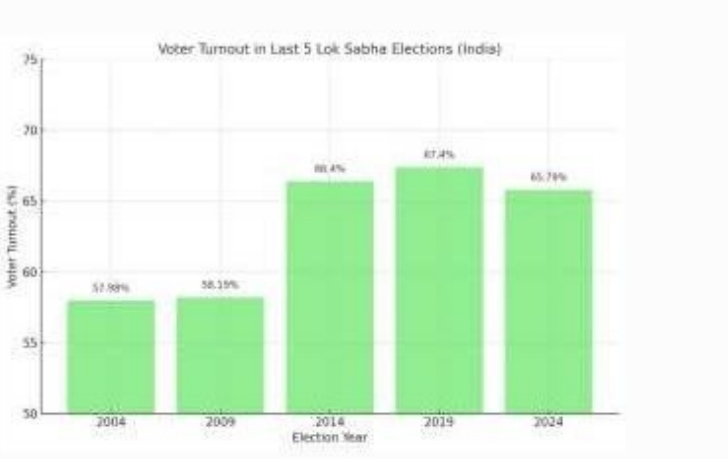
Demo Video: [Add demo video link]

Extent of AI tools: DeepFace (deep-learning face recognition) for voter authentication

Smart Vote

Impact Potential

Impact on Target Audience – The Participation Gap



~34% of voters did not vote in Lok Sabha 2024 (turnout 65.79%) – accessible, trusted online voting can close this gap.

Social

- Inclusive for elderly, differently-abled, remote & underserved citizens
- Higher participation
- Multilingual, voice-guided access

Economic & Green

- Cost-effective vs. booths and logistics
- Faster counting & results
- Paperless process, lower footprint

Trust & Governance

- Tamper-proof, auditable record
- Anonymity with public verifiability
- Transparent, accountable institutions

Stakeholders

- Voters (incl. elderly & differently-abled)
- Election authorities
- Institutions & organisations
- Public auditors & observers

- SDG 9 – Industry, Innovation & Infrastructure
- SDG 11 – Sustainable Cities & Communities
- SDG 17 – Partnerships for the Goals

Smart Vote

Feasibility and Risk Analysis

Hack for Social Cause

Feasibility of the Solution

- Working prototype built: Flask + MySQL + blockchain + face authentication
- Open-source, low-cost stack; runs in web / mobile browsers
- Cheaper than booths, staff and manual counting
- Builds on existing Aadhaar infrastructure

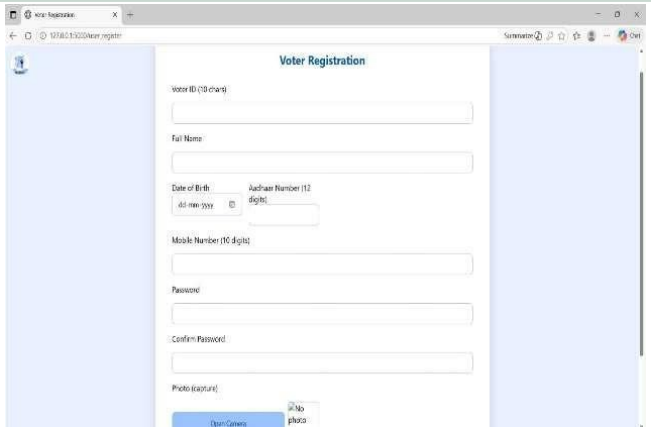
Potential Challenges & Risks

- Digital divide and low digital literacy
- Impersonation / coercion in remote voting
- Scalability for large electorates
- Legal acceptance and public trust

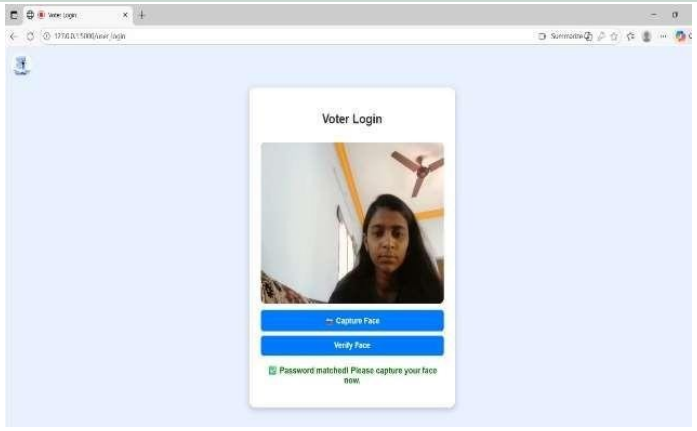
Strategies to Overcome

- Multilingual, voice-guided UI and offline registration
- Aadhaar + face verification, digital signatures, E2E encryption
- Phased rollout: pilot → institutional → large polls
- Contingency mechanisms + public blockchain audit

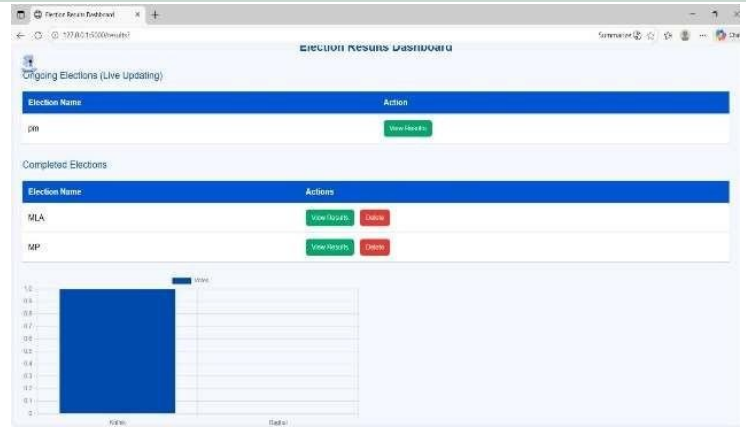
Working Prototype (Implementation)



Voter registration



Login with face verification



Election results dashboard

Team

Team Composition

Member · Role
[Member 1 name] – [Role, e.g. Blockchain / Backend / AI & Face Auth / UI & Testing]
[Member 2 name] – [Role, e.g. Blockchain / Backend / AI & Face Auth / UI & Testing]
[Member 3 name] – [Role, e.g. Blockchain / Backend / AI & Face Auth / UI & Testing]
[Member 4 name] – [Role, e.g. Blockchain / Backend / AI & Face Auth / UI & Testing]

Challenges

- [Add key challenge faced]
- [Add key challenge faced]

Our Journey

- [Add anything the team would like to share about their journey]

References

Literature Survey

Citation	Methodology / Contribution
Rivest & Smith, 2007	Voter-verifiable protocols (ThreeBallot, VAV, Twin) – less complex cryptography; voters verify independently.
Chaum 2004; Benaloh 2006	Receipt-based verifiable voting – verification while preserving anonymity.
Mercuri, 2002	Exposed weaknesses of centralized electronic counting; stressed checks and balances.
Ryan & Schneider, 2006–09	Re-encryption mix-nets for secure vote shuffling – better privacy and auditability.

References

[1] R. Rivest and W. Smith, "Three Voting Protocols: ThreeBallot, VAV, and Twin," USENIX/ACCURATE EVT Workshop, 2007.

[2] S. Popoveniuc and B. Hosp, "An introduction to secure electronic voting," IEEE Security & Privacy, vol. 10, no. 5, pp. 42–49, 2012.

[3] R. Mercuri, "Electronic vote tabulation: checks & balances," IEEE Spectrum, vol. 39, no. 10, pp. 46–50, 2002.

[4] D. Chaum, "Secret-ballot receipts: True voter-verifiable elections," IEEE Security & Privacy, vol. 2, no. 1, pp. 38–47, 2004.

[5] A. Kiayias, T. Zacharias and B. Zhang, "End-to-end verifiable elections in the standard model," EUROCRYPT, pp. 468–498, 2015.

[6] J. Benaloh, "Simple verifiable elections," 1st USENIX Workshop on Electronic Voting Technology, 2006.

[7] C. Volkamer and B. Grimm, "E-voting system usability: lessons learned from four case studies," J. Information Technology & Politics, vol. 6, no. 1, pp. 35–51, 2009.